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17 August 1988
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20 October 1978
(See 6.8)

MILITARY SPECIFICATION

MEASURING SYSTEM, SUBMARINE STORAGE BATTERY CURRENT (METRIC)

This specification is approved for use within the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a current measuring system used to measure and display the instant value and time integral of direct current (dc) in a submarine storage battery circuit.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these docuemnts shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

PPP-F-320 - Fiberboard: Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes.

MILITARY

MIL-M-14 - Molding Plastics and Molded Plastic Parts, Thermosetting.

MIL-S-901 - Shock Tests, H.I. (High Impact); Shipboard Machinery, Equipment and Systems, Requirements for.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362-5101 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 6625

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MILITARY (Continued)

MIL-E-917	- Electric Power Equipment, Basic Requirements (Naval Shipboard Use).
MIL-E-2036	- Enclosures for Electric and Electronic Equipment (Naval Shipboard).
MIL-P-15024	- Plates, Tags and Bands for Identification of Equipment.
MIL-P-15024/5	- Plates, Identification.
MIL-M-16034	- Meters, Electrical - Indicating (Switchboard and Portable Types).
MIL-M-16034/2	- Ammeters, D.C. Switchboard, 4-1/2 Inch, 250-Degree Scale.
MIL-E-17555	- Electronic and Electrical Equipment, Accessories, and Provisioned Items (Repair Parts): Packaging of.
MIL-L-19140	- Lumber and Plywood, Fire-Retardant Treated.
MIL-C-24643	- Cable and Cord Electrical, Low Smoke, for Shipboard Use, General Specification for.
MIL-C-24643/33	- Cable, Electrical, Type LS2SWU (Including Variation LS2SWUA).

STANDARDS

MILITARY

MIL-STD-167-1	- Mechanical Vibrations of Shipboard Equipment (Type I - Environmental and Type II - Internally Excited).
MIL-STD-202	- Test Methods for Electronic and Electrical Component Parts.
MIL-STD-470	- Maintainability Program for Systems and Equipment.
MIL-STD-1364	- Standard General Purpose Electronic Test Equipment.
MIL-STD-1399, Section 300	- Interface Standard for Shipboard Systems Electric Power Alternating Current. (Metric)
DOD-STD-1399, Section 304	- Interface Standard for Shipboard Systems Electrical Cables and Connectors. (Metric)

2.1.2 Other Government publications. The following other Government publication forms a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

PUBLICATION

NAVAL SEA SYSTEMS COMMAND (NAVSEA)

NAVSHIPS 0967-312-8010 - Maintainability Design Criteria Handbook for Designers (Formerly NAVSHIPS 94324) of Shipboard Electronic Equipment.

(Copies of specifications, standards, and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

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2.2 Other publication. The following document forms a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)
Y32.16 - IEEE Standard Reference Designations for Electrical and Electronics Parts and Equipments. (DoD adopted)

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.3 and 6.3).

3.2 System assemblies. The current measuring system shall consist of the following assemblies:

(a) Dc transducer. The transducer shall contain one or more iron cores, wound coils and Hall effect semiconductor devices as required to yield operation specified herein. The transducer shall provide electrical isolation between the storage battery dc busbar (rigid busbars or cables) and all circuits in the transducer. The function of the transducer is to sense magnitude and direction of dc flowing in the busbar and to send appropriate signals to the control unit.

(b) Control unit. The control unit shall utilize solid-state non-rotating parts such as transistors, other semiconductor devices, transformers, resistors, capacitors, and so forth as required to perform the required function. The function of the control unit is to process the signal derived from the transducer and to appropiately energize the display devices specified hereinafter.

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(1) Ammeters. The measuring system shall operate within the requirements of this specification with three ammeters that will be remotely located on a switchboard. Unless otherwise specified (see 6.2.1), the ammeters are not required to be furnished as a part of the measuring system. The ammeters shall be as follows:

a. Low-range. The low-range current ammeter shall be a dc ammeter of 20-0-20 ampere range conforming to MIL-M-16034 and MIL-M-16034/2.

b. Mid-range. The mid-range dc ammeter shall be a dc ammeter conforming to MIL-M-16034 and MIL-M-16034/2 in the following ranges:

500-0-500 ampere range.
800-0-800 ampere range.

c. High-range. The high-range dc ammeter shall be a dc ammeter conforming to MIL-M-16034 and MIL-M-16034/2, except the ranges shall be as follows:

4000-0-3000 ampere range.
8000-0-6000 ampere range.

(2) Ampere-hour indicator. The ampere-hour indicator shall have two digital counters, one mounted in the control unit and the other remotely mounted near the ammeters on a switchboard. The remote indicator shall be enclosed in a meter case conforming to the basic material and dimensional requirements of MIL-M-16034 and MIL-M-16034/2. The digital counter mounted in the control unit shall have a range of 0 to 9,999,999 ampere hours, and the remote digital counter shall have a range of 0 to 999,999 (see 3.4.3).

3.3 Connecting cable. The connections between the dc transducer, the control unit, and all dc meters and the ampere-hour indicators shall be made by shielded cables having a maximum resistance of 2.3 ohms per conductor per 100 meters at 25 degrees Celsius (°C) (see MIL-C-24643 and type LS2SWU of MIL-C-24643/33). Unless otherwise specified (see 6.2.1), cables shall not be furnished. Maximum cable lengths will be as follows:

(a) Between the dc transducer and the control unit - 80 meters.
(b) Between the control unit and all dc meters and the remote ampere-hour indicator:

500-500 ampere mid-range and 4000-0-3000 ampere high range: 80 meters.
800-800 ampere mid-range and 8000-0-6000 ampere high range: 160 meters.

3.3.1 Electrical connectors. Electrical connectors shall conform to section 304 of DOD-STD-1399.

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3.4 Construction of assemblies.

3.4.1 Transducer. The transducer shall conform to the requirements of MIL-E-917 and shall be constructed for bulkhead mounting. The envelope and mounting dimensions shall be as shown on figure 1. The transducer shall have a dripproof protected (45-degree) enclosure conforming to class 1 of MIL-E-2036. The transducer shall be installed by sliding the transducer over the end of the busbar. If the transducer contains one or more Hall effect semiconductor devices, replacement of these devices shall not require removal of the transducer from the busbar. The mass of the transducer shall not exceed 12 kilograms (kg).

3.4.2 Control unit. The control unit shall conform to the requirement of MIL-E-917 and shall be constructed for bulkhead mounting. Mounting and envelope dimensions shall be as shown on figure 2. The control unit shall have a drip-proof protected (45-degree) enclosure conforming to class 1 of MIL-E-2036. The mass of the control unit shall not exceed 14 kg.

3.4.3 Ampere-hour indicator. The ampere-hour indicator shall be a non-mercury type having two digital counters. Each counter shall display seven digits for the local readout and six digits for the remote. Each counter shall be operated from the control unit to indicate the time integral of the battery dc in one direction only; one counter to register the charging current ampere hours and the other to register the discharging current ampere-hours. The counter digits shall be not smaller than 3.5 millimeters (mm) high and 3 mm wide and shall be white on black background. The indicators shall be flush mounting. The bezel of the indicators shall be constructed of metal or plastic material. The metal shall be of corrosion-resistant material or material treated for corrosion resistance in accordance with MIL-E-917. The plastic material shall be in accordance with MIL-M-14 of a type specified in MIL-E-917. The dimensions of the indicators and mounting bezel shall be as shown on figure 3. A hand operable reset device shall be provided for each digital counter. The reset devices shall be mounted on the back for the counter mounted in the control unit and in the front for the counter remotely mounted. One counter shall have a legend plate designated "charge" and the other counter shall have a legend plate designated "discharge".

3.5 System connections and isolation. Positive indication or marking shall be provided on the transducer to identify the direction of charge current flow. The system assemblies shall be provided with terminal blocks or connectors for external connections. The system shall be completely isolated, both electrically and physically, from the battery busbars. Minimum values of insulation resistance and dielectric withstanding voltage shall be as shown in table I. There shall be no evidence of arcing, corona (visible, audible, or odorous) flashover, or punctured insulation following the dielectric withstanding voltage test.

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TABLE I. Insulation resistance and dielectric withstanding voltage.

Assembly	Point of measurement		Minimum insulation resistance	Minimum dielectric withstanding voltage (alternating current)
	Between	And		
Transducer	Busbar	Transducer terminals	(Megohms) 10	(Volts) 3600
	Each electrically isolated circuit	All other circuits and ground	10	1200
Indicator	Terminals	Ground	20	2600
	Terminals	Exposed metallic surfaces	20	2600
Control unit	Each electrically isolated circuit	All other circuits and ground	10	1200

3.6 System performance.

3.6.1 Accuracy. Accuracy of indication shall be as specified in table II, in terms of percent of the full scale value of the ammeters being energized for the dc indicators and in terms of percent of maximum busbar current for the low-range, mid-range, or high-range, times the time interval in hours for the ampere-hour indicator. For example, if the full scale value of the ammeter is 40 amperes (20-0-20 meters), the low-range current accuracy is plus or minus 1 percent of 40 (2 x 20 amperes) or plus or minus 0.4 amperes for any current not greater than 20 amperes. A similar example for ampere-hours involving 2 hours operation in the low current range where the accuracy is plus or minus 1.5 percent, is plus or minus 1.5 percent of 40 (2 hours x 20 amperes) or plus or minus 0.6 ampere-hour. These accuracies shall apply at reference conditions as specified in 3.6.1.1.

TABLE II. Accuracies in percentage.

Indicator	Busbar current ampere charge or discharge	
	1 to 20	20 to 8000
Low-range current ammeter	±1.0	----
Mid-range current ammeter	----	±2
High-range current ammeter	----	±2
Ampere-hour indicator	±1.5	±2.5

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3.6.1.1 Reference conditions. Accuracy requirements as shown in table II shall apply under the reference conditions shown in table III.

TABLE III. Reference conditions.

Variable	Condition
Temperature	27.5 ± 2°C
Relative humidity	30 to 60 percent
Applied frequency	60 ± 1 hertz
Applied voltage	115 ± 1 volts
Operating position	On vertical panel
Warm-up time	Reading time

3.6.2 Input power (control unit). Circuits of the control unit will be connected to and shall operate from a 60 hertz, single phase source of 115 volts type I power in accordance with section 300 of MIL-STD-1399. The indications given by the meters and the indicators shall not be affected to an extent greater than that specified in 3.9 by the steady-state variations in input voltage and frequency of the magnitude given by section 300 of MIL-STD-1399. System calibration shall not be permanently affected by voltage excursions from 103.5 to 126.5 volts. The power required from the 60 hertz source to operate the system shall be not more than 100 watts under all operating conditions specified herein.

3.6.3 Busbar current. When busbar current exceeding the end-scale value of the ammeters (see 3.2(b)(1)) flows, the control unit circuits shall provide limiting of the energy applied to these ammeters so that while the pointer is held at or beyond the endscale point the indicator is not mechanically or thermally stressed at levels above those which can be withstood continuously. The system shall continuously operate at 8000-ampere busbar current. Fault currents up to 20,000 amperes for 1 second shall cause no damage to any assemblies, subassemblies and parts of the system. No damage shall result from rates of change of busbar current up to 2,000,000 amperes per second.

3.6.4 Busbar voltage. The normal busbar to ground dc voltage is 270 volts with a range of 175 to 355 volts with one battery of 126 or 128 cells.

3.6.5 Calibration. Means shall be provided for the calibration of the measuring system and shall be located within the control unit. Length of the cables between the control unit, the transducer, the ammeters, and indicator may vary from installation to installation (see 3.3). The adjustment device settings shall be secure from change caused by shock, vibration, and other environmental conditions specified herein (see 3.7).

3.7 Environmental requirements.

3.7.1 Mechanical.

(a) Shock. System assemblies, subassemblies, and parts shall meet the requirements for type A, grade A, hull (bulkhead) mounted, lightweight, class I equipment as specified in MIL-S-901. Change in indication of the meters and the indicator (following

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reset of the meter pointers to zero) shall not exceed the value specified in 3.9. Following the shock test of 4.7.2.4, the system shall show no signs of breakage or distortion of parts sufficient to impair operation. Insulation resistance shall be not lower than the values of table I and dielectric withstanding voltage shall be not lower than 65 percent of the values of table I.

(b) Vibration. System parts shall withstand the type I vibration requirements of MIL-STD-167-1 without change in indication of the meters and the indicator (following reset of the meter pointers to zero) exceeding the value specified in 3.9. Following the vibration test of 4.7.2.5, the system shall show no signs of breakage or distortion of parts sufficient to impair operation. Insulation resistance shall be not lower than the values of table I.

3.7.2 Temperature range. The system shall operate in the temperature range of 5 to 65°C. Reference temperature of initial calibration shall be 27.5 ± 2°C for the meters, indicator, control unit and the transducer. Change in indication due to ambient temperature excursions of 37.5 above or 22.5°C below the reference temperature shall not exceed the values specified in 3.9.

3.7.3 Relative humidity. The transducer ampere-hour indicator and control unit shall be able to be stored for 168 hours at 100 percent relative humidity. Following such storage, insulation resistance shall be not less than 2 megohms, and the transducer ampere-hour indicator and control unit shall show no signs of material damage or corrosion.

3.8 Inclination. When inclined to an angle of 60 degrees from the vertical in any direction, the system shall not exceed the errors specified in 3.9.

3.9 Errors due to environmental conditions. In addition to the errors permitted by table II, additional errors as shown in table IV are permitted under or following exposure to specified environmental conditions.

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TABLE IV. Errors in percentage of full scale meter value due to environmental conditions.

Condition	Busbar current, amperes 1 to 8000
Temperature range (3.7.2)	
Ammeters	±1 1/
Ampere-hour indicator	
Input power (3.6.2)	
Ammeters	1/
Ampere-hour indicator	1/
Vibration (3.7.1(b))	
Ammeters (following zero reset)	±2
Ampere-hour indicator	2/
Shock (3.7.1(a))	
Ammeters (following zero reset)	±5
Ampere-hour indicator	2/
Inclination (3.8)	
Ammeters	±2
Ampere-hour indicator	No effect

1/ Included in table II values (see 3.6.1).
2/ Count change not greater than minus 0 plus 10 ampere-hours.

3.10 Recovered materials. Unless otherwise specified herein, all equipment, material, and articles incorporated in the products covered by this specification shall be new and may be fabricated using materials produced from recovered materials to the maximum extent practicable without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that use of used or rebuilt products is allowed under this specification unless otherwise specifically specified.

3.11 Drawings. When specified in the contract or order, drawings shall be prepared (see 6.2.2 and appendix A).

3.12 Maintainability. The equipment shall be constructed so that it can be maintained by easy replacement of removable parts or modules. The design shall conform to the design-for-maintainability requirements of MIL-STD-470 and NAVSHIPS 0967-312-8010. The equipment repair time shall be 1 hour.

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3.12.1 Test equipment. The system shall be constructed so that it can be maintained, aligned, calibrated, and repaired with the aid of the following test equipment selected from MIL-STD-1364.

(a) Clamp-on V ammeter, 0 to 600 amperes and 0 to 600 Vac.
(b) Multimeter, 0 to 400 Vdc, 0 to 800 Vac, 0 to 16 ampere dc and 0 to 200 megohm range.
(c) Oscilloscope.
(d) Dc milliammeter, 0 to 150 milliampere (mA) range.
(e) Ac voltmeter, 0 to 600 volts range.

3.12.1.1 A contractor desiring to utilize test equipment other than the ranges specified herein shall submit the list to NAVSEA for approval with justification of the necessity for using such test equipment before the start of manufacture of the system.

3.12.2 Repair of removable parts and subassemblies. The repairability of any removable modules, parts or subassemblies shall be determined in accordance with MIL-E-917. The time required to repair any malfunctioned parts, module or subassembly, after removal from the system, shall be not more than 10 minutes.

3.13 Reliability. The current measuring system shall have a mean-time-between-failure of not less than 10,000 hours. Definition of failure shall be increase of errors of indication beyond the limits shown in tables II and III or failure to indicate.

3.14 Identification of equipment division by reference designation. To provide identification that will aid documentation and to facilitate maintenance, each part, subassembly or assembly of the current measuring system shall be identified by reference designations in accordance with ANSI Y32.16.

3.15 Identification plate. The current measuring system shall be equipped with an identification plate in accordance with MIL-P-15024. The identification plates shall conform to normal marking requirements of MIL-P-15024/5.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve

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the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Inspection system. When specified in the contract or order an inspection system program plan shall be prepared (see 6.2.2).

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 First article inspection. First article inspection shall consist of the examination and tests specified in table V.

TABLE V. Inspections.

Inspection	First article inspection	Quality conformance inspection	Requirement	Inspection
General examination	X	X	3.4	4.7.1
Initial accuracy	X	X	3.6.1	4.7.2.1
Input power variation	X	X	3.6.2	4.7.2.2
Sustained high current, and short circuit	X		3.6.3	4.7.2.3
Shock	X		3.7.1(a)	4.7.2.4
Vibration	X		3.7.1(b)	4.7.2.5
Inclination	X		3.8	4.7.2.7
Temperature cycling	X		3.7.3	4.7.2.6
Insulation resistance	X	X	Table I	4.7.3
Humidity	X		3.7.3	4.7.3.2
Dielectric withstanding voltage	X	X	Table I	4.7.4
Temperature rise (control unit only)	X		3.6.2	4.7.5
Calibration	X	X	3.6.5	4.7.6
Maintainability demonstration	X		3.12	4.8
Enclosure	X		3.4.1 and 3.4.2	4.9

4.4 Quality conformance inspection. Each current measuring system offered for delivery shall be subjected to the quality conformance inspection as specified in table V.

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4.5 Test conditions. Unless otherwise specified, all tests shall be conducted under the reference conditions as specified in 3.6.1.1. With the exception of temperature cycling test (see 4.7.2.6), the tests may be conducted at ambient temperature of 25 ± 2°C and corrections to 27.5 ± 2°C (reference conditions) shall be made to all readings and measurements. The specific ambient temperature at the time of test shall be measured and recorded. Necessary adjustment and calibration of the system shall be made before the tests are started; no adjustments or calibration shall be allowed during or between the tests.

4.6 Test plan and test report. When specified in the contract or order, a test plan and test report shall be prepared (see 6.2.2 and appendix B).

4.7 Inspection procedure.

4.7.1 Visual examination. Units and assemblies of the current measuring system shall be visually examined to determine that:

(a) Parts used are as identified by drawings and fabrication techniques.
(b) Materials and fabrication techniques used in construction of parts such as the transducer, transformer, reactors and other fabricated wound magnetic parts are as identified by drawings. This examination shall include but may not be limited to verification of use of:

(1) Correct grade and dimensions of core iron.
(2) Correct core (or ground), winding, layer and outer wrapper insulation material.
(3) Correct gauge, insulation and lead termination of magnet wire.
(4) Correct choice of an application of insulation varnish or other impregnant.
(5) Correct marking of leads. (Marking on parts is consistent with schematic and connection diagrams.)

(c) Mounting of parts is as shown on drawings. This includes examinations for correct location, orientation, securing method (fasteners, insulators, and others) and, where applicable, torquing.
(d) Marking of parts (or, where parts are too small to mark, marking of adjacent panel) and terminals is complete and consistent with schematic and connection diagrams.
(e) Hookup wire is as shown on drawings.
(f) Hookup wire is neatly formed into groups and laced, tied or clamped securely into place (with nonflammable cord or clamps) to prevent chafing caused by vibration.
(g) Wire ends are marked.
(h) Grommets are provided where wires pass through holes in metal panels.
(i) Wires are not bent sharply around corners.
(j) Wires are connected by solder or other mechanical means (with locking devices).

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(k) Clearances called for on drawings (see appendix A, 30.1 (h)) for part removal (parts need not actually be removed) or access of tools or test equipment probes or leads are adequate.
(l) Indicator mounting panel drilling plan and mounting means as specified.
(m) Soldering joints are properly done.

4.7.2 Accuracy determination.

4.7.2.1 Initial. The current measuring system shall be operated at reference condition as specified in 3.6.1.1 and readings made at busbar current (or equivalent ampere-turns) and ampere-hour values of table VI.

TABLE VI. Initial accuracy.

Charge busbar current (amperes)	Discharge busbar current (amperes)	Ampere-hours (minimum)
1	1	3
5 1/	5 1/	2.5 1/
50 1/	50 1/	5 1/
250	250	125
500 1/	500 1/	250
800	800	250
1000	1000	50 1/
2500	2500	250
5000 1/	5000 1/	100 1/
6000 1/	8000 1/	100 1/

1/ Units after first article tested to these values only.

4.7.2.2 Input power variations. Input voltage and frequency shall be adjusted to the following values to check the system accuracy at 5, 500, and 800 charge amperes and 500 and 800 discharge amperes busbar current (use table VI for the minimum ampere-hours required for each value of current):

(a) 103.5 volts at 63 hertz.
(b) 103.5 volts at 57 hertz.
(c) 126.5 volts at 63 hertz.
(d) 126.5 volts at 57 hertz.

4.7.2.2.1 If data shows that errors are equal to or less than 2/3 of the sum of the applicable values from tables II and IV, no additional values of busbar current need be checked. Otherwise, each value of table VI shall be checked. Voltage shall also be adjusted to 103.5 volts at 60 hertz, then 126.5 volts at 60 hertz for 15 minutes each, then indication at 50 amperes of busbar current checked.

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4.7.2.3 Sustained high current fault current. The accuracy test of 4.7.2.2 shall be made after the system has been operated at currents (of equivalent ampere-turn) for the periods as follows:

(a) 8000 amperes for 10 minutes.
(b) 20,000 amperes for 1 second.

4.7.2.4 Shock. The transducer, control unit and ampere-hour indicator shall be mounted on a fixture fabricated to simulate actual installation circumstances. The transducer shall be wholly supported by the normal mounting means and not be furnished any support by the busbar. The ampere-hour indicator shall be mounted on a portion of the fixture similar to fixture 6D-1 of figure 7-1 of MIL-S-901. The fixture, with the system parts shall be mounted on a lightweight shock test machine and tested as specified in MIL-S-901. Tests shall be performed with approximately 50 amperes of busbar current flowing.

4.7.2.4.1 Definition of failure of shock test. Following the shock test of 4.7.2.4, failures shall be defined as follows:

(a) Failure to indicate or errors of indication exceeding the values of table IV as determined by the accuracy checks of 4.7.2.2.
(b) Breakage or distortion of any parts sufficient to impair operation.
(c) Insulation resistance lower than the values of table I.
(d) Dielectric withstanding voltage lower than 65 percent of the specified values of table I.

4.7.2.5 Vibration. Using the fixture as specified in 4.7.2.4, the system assemblies and subassemblies shall be subjected to a type I test of MIL-STD-167-1. Failure of this test shall be determined by the same criteria as in 4.7.2.4.1, except that dielectric withstanding voltage need not be determined.

4.7.2.6 Temperature cycling. A current measuring system shall be subjected to temperature cycles over the range of temperatures of 3.7.2 as follows and the test shall be performed to check the system accuracy in accordance with 4.7.2.2:

(a) Determine accuracy at reference conditions.
(b) Condition the system at minimum temperature (5°C) for 8 hours.
(c) Determine accuracy at the minimum temperature (5°C).
(d) Condition the system at reference temperature 27.5 $\pm$ 2°C for 8 hours.
(e) Repeat (b), (c), and (d) twice for a total of three cycles to minimum temperature and return to reference temperature.
(f) Condition the system at maximum temperature (65°C) for 8 hours.
(g) Determine accuracy at maximum temperature (65°C).
(h) Condition the system at reference temperature 27.5 $\pm$ 2°C for 8 hours.
(i) Repeat (f), (g), and (h) twice for a total of three cycles to maximum temperature and return to reference temperature.
(j) Maximum change in indication between observations at minimum and maximum temperatures shall be considered to be the effect of temperature.

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4.7.2.7 Inclination. Using the fixture as specified in 4.7.2.4, the system shall be inclined forward, backward, to the right, and to the left to an angle of 60 degrees from the vertical to test conformance to accuracy requirement in table IV. The accuracy shall be checked at busbar current and ampere-hours reading as follows:

Charge busbar current (Amperes)	Discharge busbar current (Amperes)	Ampere-hours (Minimum)
50	50	5

4.7.3 Insulation resistance. Insulation resistance test shall be conducted in accordance with method 302 of MIL-STD-202. Test conditions shall be as follows:

(a) Test potential: 500 volts ± 10 percent dc.
(b) Points of measurement: As specified in table I.
(c) Electrification time: 60 seconds minimum.

4.7.3.1 Isolated circuits. Electrically isolated circuits shall be determined by application of the following criteria:

(a) Circuits whose only connection to each other is by electromagnetic coupling through a magnetic core that is shared in common by the circuit shall be considered to be electrically isolated from each other.
(b) Circuits where only connection to each other is through a capacitor shall not be considered to be electrically isolated from each other. When the purpose is to test circuits internal to the current measuring system and only then, such circuits shall be temporarily interconnected with a jumper wire or test lead during the test.

4.7.3.2 Humidity. A transducer ampere-hour indicator and a control unit shall be stored for 168 hours at 100 percent relative humidity at 31 ± 2°C. Upon completion of this period, the transducer ampere-hour indicator and control unit shall be tested as specified in 4.7.3 for conformance to 3.7.3. The test shall be performed within 5 minutes after completion of the humidity period. Following the test, the transducer ampere-hour indicator and control unit shall be carefully examined for any signs of material damage and corrosion.

4.7.4 Dielectric withstanding voltage. Dielectric withstanding voltage shall be conducted in accordance with method 301 of MIL-STD-202. Test conditions shall be as follows:

(a) Magnitude of test voltages shall be as specified in table I.
(b) Nature of test voltage. The test voltage shall approximate a true sine wave of a frequency not less than the rated frequency of the circuit under test.
(c) Points of measurement. Points of measurement shall be as specified in table I. (Electrically isolated circuits shall be as defined in 4.7.3.1.)

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(d) Duration of test. Duration shall be 60 seconds, minimum. The test voltage shall be removed by adjustment of its value to zero, not by sudden interruption.
(e) Definition of failure. Any evidence of arcing, corona (visible or odorous), flashover or punctured insulation shall be interpreted as failure of the test.
(f) Dielectric withstanding voltage test shall be conducted after all other tests have been completed.

4.7.5 Temperature rise (control unit only). The control unit shall be operated with meters and indicator or simulated impedances connected and also with a real or simulated transducer connected. The transducer (or simulator) energy shall be varied to determine the value (within limits corresponding to busbar currents from 0 to 8000 amperes) resulting in maximum power dissipation in the control unit circuits. The ambient temperatures shall be maintained at 65°C. Measurement of temperatures of circuit parts shall be performed as specified in MIL-E-917. The power consumed by the control unit shall be determined to conform to 3.6.2.

4.7.6 Calibration demonstration. Resistance shall be inserted between the control unit and the transducer and between the control unit and the meters and the indicator to simulate various cable lengths. The resistance shall be adjusted to 0.2, 0.5, zero, 1 and 3.5 ohms for each conductor and conformance to 3.6.5 shall be demonstrated.

4.8 Maintainability demonstration test. The times required to accomplish normal corrective maintenance and preventive maintenance tasks shall be demonstrated. The elapsed time and man-hours required to replace each of the on board repair parts shall be reported. The elapsed time and man-hours to perform each proposed preventive maintenance task shall also be reported. These time recordings may be obtained where possible from the maintenance actions associated with other tests required by this specification in order to minimize the amount of special effort required to obtain the maintainability data. The time data shall include the fault correction time for corrective maintenance tasks and servicing time for preventive maintenance exclusive of any subsequent test time. The procedure for the maintainability demonstration shall be the same as the procedures that will be in the equipment technical manual. When specified in the contract or order, a maintainability demonstration report shall be prepared (see 6.2.2).

4.9 Enclosures. The enclosures shall be tested to determine conformance to 3.4.1 and 3.4.2.

4.10 Inspection of packaging. Sample packs, and the inspection of the preservation, packing and marking for shipment, stowage, and storage shall be in accordance with the requirements of section 5 and the documents specified therein.

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5. PACKAGING

(The packaging requirements specified herein apply only for direct Government acquisition. For the extent of applicability of the packaging requirements of referenced documents listed in section 2, see 6.4.)

5.1 General.

5.1.1 Navy shipboard stowage fire-retardant requirements.

(a) Treated lumber and plywood. Unless otherwise specified (see 6.2.1), all lumber and plywood including laminated veneer material used in shipping containers and pallet construction, members, blocking, bracing, and reinforcing shall be fire-retardant treated material conforming to MIL-L-19140 as follows:

Levels A and B - Type II - weather resistant.
Category 1 - general use.
Level C - Type I - non-weather resistant.
Category 1 - general use.

(b) Fiberboard. Fiberboard used in the construction of class-domestic, non-weather resistant fiberboard, cleated fiberboard boxes including interior packaging forms shall meet the flame spread index and the specific optic density requirements of PPP-F-320 and amendments thereto.

(c) Cushioning and wrapping materials. The use of excelsior, newspaper, shredded paper (all types), and similar hyroscopic or nonneutral materials and all types of loose fill materials for packaging applications such as cushioning, fill, stuffing, and dunnage is prohibited. Materials selected for cushioning and wrapping shall have properties (characteristics) for resistance to fire (see 6.5). Cushioning or wrapping materials, as applicable, shall be provided to prevent item and package damage and to prevent free movement of the container contents.

5.2 Packing requirements. System assemblies shall be preserved level A, C or commercial, packed level A, B, C, or commercial, and marked in accordance with MIL-E-17555, as specified (see 6.2.1).

6. NOTES

6.1 Intended use. The measuring system covered by this specification measures the instant value and time integral of dc in submarine storage battery circuits.

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6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) If first article is required (see 3.1).
(c) If ammeters should be furnished (see 3.2(b)(1)).
(d) When cable is required (see 3.3).
(e) Fire-retardant material requirements (see 5.1.1).
(f) Preservation, packing, and marking requirements (see 5.2).

6.2.2 Data requirements. When this specification is used in an acquisition and data are required to be delivered, the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (CDRL), incorporated into the contract. When the provisions of DoD FAR Supplement, Part 27, Sub-Part 27.475-1 (DD Form 1423) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification are cited in the following paragraphs.

Paragraph no.	Data requirement title	Applicable DID no.	Option
3.11 and appendix A	Engineering drawings	DI-DRPR-80651	----
4.1.2	Inspection system program plan	DI-R-4803	----
4.6 and appendix B	Test reports	DI-MISC-80653	----
4.8	Report, maintainability demonstration	DI-R-7113	----

(Data item descriptions related to this specification, and identified in section 6 will be approved and listed as such in DoD 5010.12-L., AMSDL. Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.2.2.1 The data requirements of 6.2.2 and any task in sections 3, 4, or 5 of this specification required to be performed to meet a data requirement may be waived by the contracting/acquisition activity upon certification by the offeror that identical data were submitted by the offeror and accepted by the Government under a previous contract for identical item acquired to this specification. This does not apply to specific data which may be required for each contract regardless of whether an identical item has been supplied previously (for example, test reports).

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6.3 First article. When a first article inspection is required, the item should be a first article sample. The first article should consist of one unit. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.4 Sub-contracted material and parts. The packaging requirements of referenced documents listed in section 2 do not apply when material and parts are acquired by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

6.5 Cushioning and wrapping materials (see 5.1.1)(c)). Materials having properties for resistance to fire and acceptable for use within unit packs and shipping containers for Navy shipboard stowage furniture item(s) acquisitions are:

Material	Specification
Paper, kraft treated (fire-resistant)	A-A-1894
Paper, kraft, wrapping	UU-P-286, type II grade C or D
Fiberboard	PPP-F-320, class-domestic/fire-retardant
Plastic film, flexible cellular	PPP-C-795, class 3 - fire-retardant
Polystyrene expanded, resilient	PPP-C-850, grade SE
Plastic, open cell, cushioning	PPP-C-1842, type I, style B
Bound fiber	PPP-C-1120, type III or IV, class C
Rubber, latex foam	MIL-R-5001, grade A
Rubber, cellular	MIL-R-6130, grade A
Fibrous glass	MIL-C-17435
Polystyrene foam	MIL-P-19644, type II
Rubber, cellular, synthetic	MIL-R-20092, class 5
Polyurethane foam	MIL-P-26514
Polyurethane foam Flexible, open cell	MIL-F-81334
Foam, combustion retardant, for cushioning supply items aboard Navy ships	MIL-F-87090

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6.6 Provisioning. Provisioning Technical Documentation (PTD), spare parts, and repair parts should be furnished as specified in the contract.

6.6.1 When ordering spare parts or repair parts for the equipment covered by this specification, the contract should state that such spare parts and repair parts should meet the same requirements and quality assurance provisions as the parts used in the manufacture of the equipment. Packaging for such parts should also be specified.

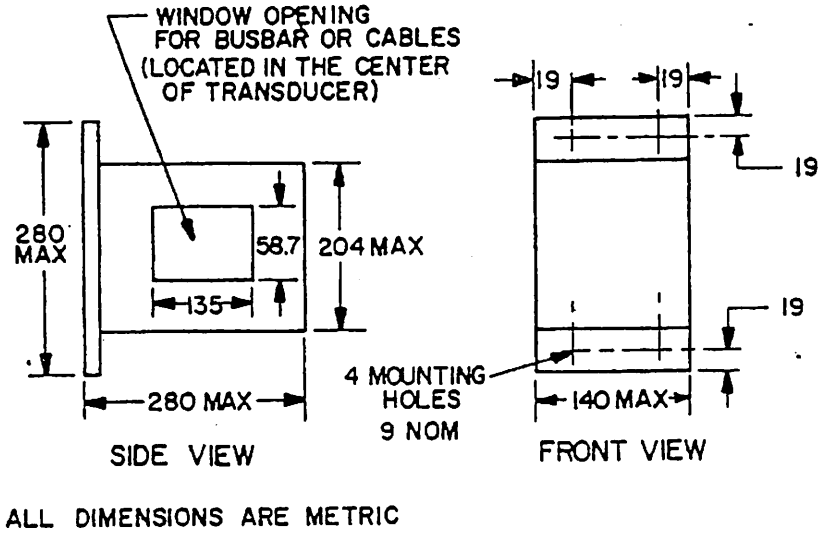
6.7 Subject term (key word) listing.

Ampere-hour indicator
Busbar current
Control unit
Dielectric withstanding voltage
Insulation resistance
Transducer

6.8 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

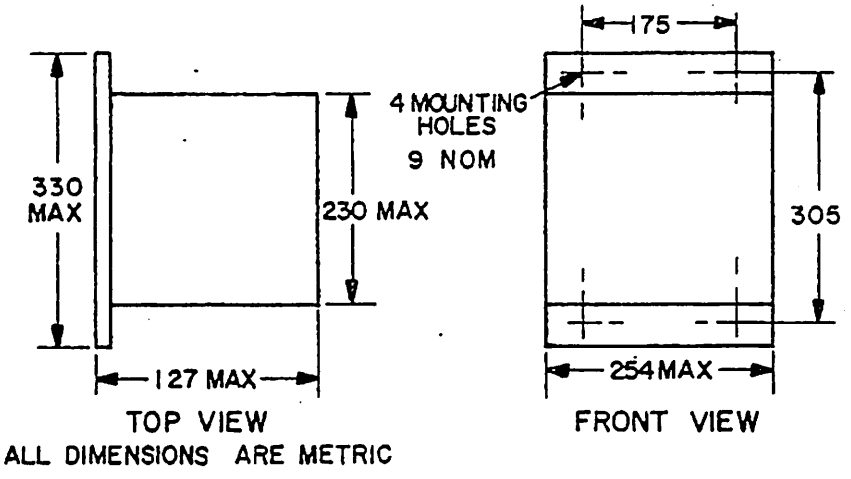
Preparing activity:
Navy - SH
(Project 6625-N913)

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SH 10655A

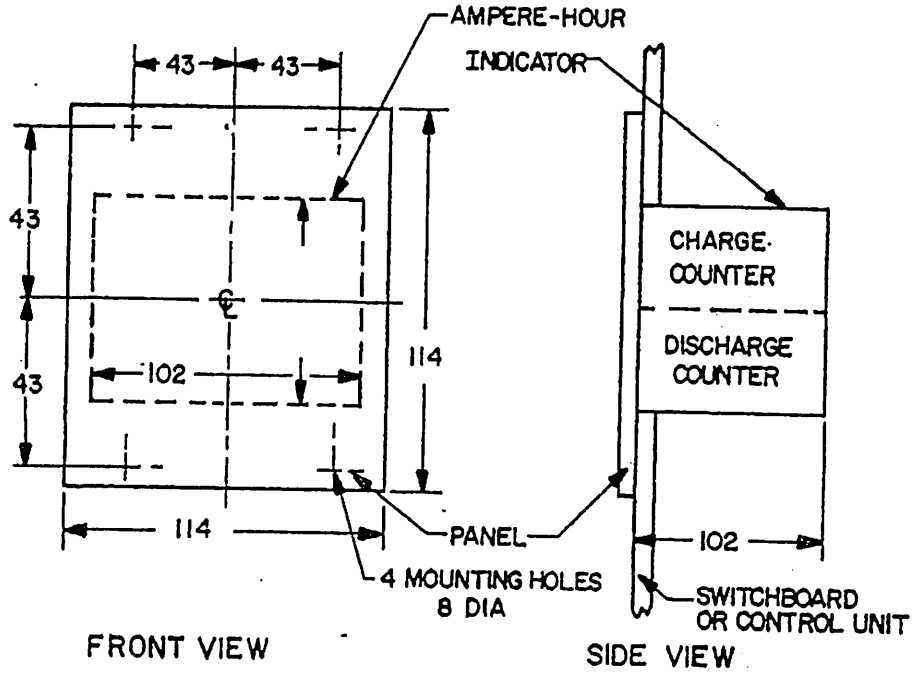
FIGURE 1. Dimension of bulkhead mounted transducer.



SH 10656A

FIGURE 2. Dimensions of control unit.

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SH 10657A

FIGURE 3. Dimensions of ampere-hour indicator.

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APPENDIX A

ENGINEERING DRAWINGS TECHNICAL CONTENT REQUIREMENTS

10. SCOPE

10.1 Scope. This appendix covers the technical content requirements that shall be in included on drawings when required by the contract or order. This appendix is mandatory only when DI-DRPR-80651 is cited on the DD Form 1423.

20. APPLICABLE DOCUMENTS

20.1 Government documents.

20.1.1 Specification and standard. The following specification and standard form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATION

MILITARY

DOD-D-1000 - Drawings, Engineering and Associated Lists. (Metric)

STANDARD

MILITARY

DOD-STD-100 - Engineering Drawing Practices. (Metric)

(Copies of specifications and standards required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

30. DRAWINGS

30.1 Drawings and certification data. When required by the contract or order, drawings shall contain, in addition to that required by DOD-STD-100 and DOD-D-1000, the following information:

(a) Manufacturer's name and model number.
(b) Reference to this specification.
(c) List of parts, tabulating part identification, specification data of part, find number corresponding to part flags of outline drawings.
(d) Core and insulation system, including varnish impregnation of magnetic parts.
(e) Identification and stress level data (part manufacturer's and inuse ratings) of each electrical part.

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(f) Schematic diagrams.
(g) Shock classification and approval reference.
(h) Outline drawing of each assembly including:

(1) Overall and mounting dimensions.
(2) Location and size of mounting holes.
(3) Mass of each assembly.
(4) Panel cutout data for the indicator.
(5) Position, size and marking of terminals used for power input and system part interconnection.
(6) Dissassembly and reassembly sketches (for example, an exploded view) required for transducer installation on a busbar.
(7) Clearance requirements for ventilation, part removal, and test equipment access.

(i) Interconnection cable requirements such as permissible resistance, and shielding requirements, and so forth.
(j) Amperes, and power required from the 115 volt, 60 hertz source.
(k) Graphic or tabular presentation of characteristics of transducer, control unit circuits, meters, and indicator. This data is required for troubleshooting assistance and need not contain complete calibration procedures and instructions. Normal readings at relevant circuit points under specified condition shall be presented.
(l) List of recommended test equipment such as multimeters or portable standard instruments.
(m) Schematic circuit of the complete system.
(n) Connection diagram showing approximate part location, cable routing within equipment, and terminal and wire marking.
(o) Description of operation, describing briefly how the circuit parts perform their functions to yield specified systems operation.

Drawings shall be not smaller than 430 by 560 millimeters (mm) nor larger than 710 by 1020 mm. The number of sheets required for the drawings scaled to size shall be the minimum consistent with contents and clarity.

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APPENDIX B

TEST REPORT TECHNICAL CONTENT REQUIREMENTS

10. SCOPE

10.1 Scope. This appendix covers the technical requirements that shall be included in quality conformance inspection reports when required by the contract or order. This appendix is mandatory only when DI-MISC-80653 is cited on the DD Form 1423.

20. APPLICABLE DOCUMENTS

This section is not applicable to this appendix.

30. Test plan and test report.

30.1 Test plan. The test plan shall include but not be limited to detailed test procedures of all tests specified herein, list of test equipment, list of instrumentation, environmental conditions, required data, and other pertinent information as required by each test.

30.2 Test report. The test report shall address each test specified herein and shall include but not be limited to the following items:

(a) Test article, identification or full description of units or equipments tested.
(b) Data and geographical location of test or evaluation.
(c) Statement of test/evaluation objectives, including type, unit of measure, and quantitative goals and requirements to be demonstrated.
(d) Discussion of methods and conditions of test, including methods of evaluating the data obtained and comparison of the conditions with those anticipated in ultimate deployment and use of contract item.
(e) Results obtained, including specific identification and discussion of objectives demonstrated satisfactorily and identification of problems encountered and their solutions.
(f) Failure analysis and corrective action. Total operating time prior to occurrence of failure.
(g) Principal environmental stress conditions, operating modes, operating time, and cycling conditions.
(h) Contractor's conclusions and recommendations based on evaluation of test results.

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